Novia One.

FMP · MODULE 4 · IN-DEPTH

OCCUPATIONAL CERTIFICATE · FINANCIAL MARKETS PRACTITIONER

Equity Securities, explained.

A full study workbook written the way your lecturer would teach it — building each idea from the ground up, with worked examples, the formulas you'll be tested on, exam-trap warnings, and a self-test at the end of every chapter.

HOW TO USE THIS WORKBOOK

Read it like a set of lecture notes, not a glossary. Each section opens with the *why*, then introduces the terms, then shows you how they behave. Three kinds of boxes recur throughout:

Definition — the precise wording. Learn these verbatim; the exam rewards exact terminology.

LECTURER'S NOTE — the intuition, the analogy, the "why this matters" that makes it stick.

EXAM WATCH — the specific way this topic is twisted into a wrong answer. Your exam uses **three-option** MCQs, and the wrong options are usually a made-up term, the opposite of the truth, or a neighbouring role-player. Each "Exam watch" flags exactly that.

TIP Pair this workbook with the **Summarised Reference** (the <10-page version) for last-minute revision; this one is for understanding, that one is for recall.

Chapter 1 — Introduction to Equity Securities

Learning goal: understand what equity *is*, the rights it confers, the types that exist, and how we think about its risk, return and value.

1.1 What equity capital actually is

Start with the single sentence everything else hangs off: **equity capital represents the share capital of a company**. When you buy a share, you are not lending the company money — you are buying a slice of *ownership*. That distinction drives every difference between equity and debt we'll meet in this module.

> **Capital structure** refers to the amount of **debt and/or equity** a firm employs to fund its operations and finance its assets.

A company needs money to operate and grow. It can raise that money two ways: borrow it (debt) or sell ownership (equity). There are real **trade-offs**: debt must be repaid with interest but doesn't dilute ownership; equity never has to be repaid but hands over a piece of the company and its profits. Managers balance the two to find the **optimal capital structure**.

> **LECTURER'S NOTE** Think of debt as a tenant who pays you rent but can evict-by-default if you miss payments, and equity as taking on a co-owner who shares the upside and the downside. Neither is "better" — the art is the mix.

1.2 The four rights of a shareholder

Because a share is ownership, it carries ownership rights. Memorise these four — they are the backbone of the whole topic:

- **Share in the company's profits** — typically through dividends.
- **Share in the assets of the company if it goes into liquidation** — the residual claim (you get what's left after creditors).
- **Appoint the directors** of the company.
- **Vote at shareholder meetings.**

1.3 Why invest in equities — and why not

Companies raise external capital either by borrowing or by **issuing (selling) equity securities**. Issuing shares is a company's main way of raising equity capital. From the investor's side, here is the honest balance sheet of owning shares:

Advantages

- **Great long-term returns** — historically beat cash, bonds and property. From 2001–2018 SA equity outperformed SA cash and bonds on **11 occasions**.
- **Global exposure** — multinationals let you grow with the world economy.
- **Inflation proofing** — over long periods equities repeatedly outpace cash.
- **Income** — dividend payers give a growing income (but companies are **not always obliged** to pay).
- **Tax & estate planning** — TFSA (up to R30k), endowment, or retirement (up to R350k / 27.5% of income).

Disadvantages & horizon

- **Volatility** — factors outside the company's control move the price; it can fall below cost.
- Treat equity as a **10-year** view; recovery needs time.
- If you need the money in **under 5 years**, the stock market is probably not for you.

Who & when

- **Who:** people with a long time horizon (10 years+).
- **When:** during inflation, increasing economic activity, and low interest rates.

1.4 The role of equity in the economy

Zoom out from the individual investor. Equity markets do four jobs for the economy as a whole:

Capital raising	Companies issue equity to fund expansion, R&D and other activities.
Ownership alignment	Aligns shareholders' interests with management — promotes good governance.
Investor participation	Lets individuals and institutions share in a company's success and growth.
Risk distribution	Spreads risk across many investors — contributing to market stability.

1.5 Types of equity securities

Equity securities represent the shareholder's stake in a company — mostly as **common** and **preferred** stock. Investors use them to profit from a company's future growth, so you must be able to tell them apart.

(a) Common (ordinary) equity

Common shares are the **predominant** type of equity security. Their characteristics:

- **Ownership interest** — you share in the operating performance of the company.
- **Residual claimant** — your claim on net assets ranks *last*, after all liabilities are paid.
- **Governance participation** — voting rights on major decisions.
- **Share in the profits** — the company may pay some or all of its net income as cash dividends, but is **not obligated** to.

Voting rights — statutory vs cumulative

Voting lets shareholders participate in major corporate governance decisions: **electing board members, deciding on a merger or takeover, and selecting outside auditors**. Shareholders who can't attend the annual meeting may vote by **proxy** — appointing another shareholder, a representative, or management to vote on their behalf.

Statutory voting

One vote per share, applied to each director **separately**. With 100 shares you cast up to 100 votes for each board seat — but no more than 100 for any one director.

Cumulative voting

You may **concentrate** your total votes on chosen candidates rather than spreading them evenly. This protects **minority** shareholders — they can pool votes to get a director elected.

WORKED EXAMPLE Four directors are to be elected and you own 100 shares. **Statutory:** up to 100 votes per director (max 100 each). **Cumulative:** 4 directors × 100 shares = **400 votes**, which you can split in any proportion — e.g. all 400 onto one candidate you care about.

Two special common-share forms

- **Callable common shares** — the company can buy them back at a preset price.
- **Putable common shares** — the holder can sell them back to the company at a preset price.

(b) Preferred (preference) equity

- Like the interest on debt, preference **dividends are fixed** and generally **higher** than common dividends.
- But **unlike interest, preference dividends are not a contractual obligation** of the company.
- Like common shares, preferreds can be **perpetual** (no fixed maturity), can pay dividends indefinitely, and can be **callable or putable**.

Preference dividends come in four base flavours, which combine:

FLAVOUR	WHAT IT MEANS
Cumulative	Missed (unpaid) dividends accumulate and must be paid before common dividends.
Non-cumulative	Missed dividends are simply lost — they do not accrue.
Participating	Get the standard dividend **plus** an extra dividend if profits exceed a set level; may also share in extra assets on liquidation. Common for smaller, riskier companies.
Non-participating	Receive only the fixed dividend and the par value of the shares on liquidation.

These combine: e.g. cumulative participating, cumulative non-participating, non-cumulative participating, non-cumulative non-participating.

Convertible preference shares

These **entitle the holder to convert** their preference shares into a specified number of common shares; the **conversion ratio is fixed at issuance**. They are a popular financing tool in **venture capital and private equity** because the issuing companies are higher-risk and it may be a long time before they list and issue common shares to the public.

Why investors like convertibles

- A **higher dividend** than they'd earn on the company's common shares.
- The opportunity to **share in profits**.
- Upside: benefit from a **rise in the common share price** via the conversion option.
- **Lower price volatility** than the underlying common shares, because the dividend is known and stable.

(c) Private equity & (d) public equity

Private equity is ownership in companies that are not publicly traded — think venture capital and leveraged buyouts. **Public equity** is shares listed and traded on an exchange.

> **Leveraged buyout (LBO)** A group of investors, **primarily using debt**, buys all the outstanding common shares of a publicly traded company. If the buyers are the company's **existing management**, it's a **management buyout (MBO)**. After purchase the investors take full control and the company is **taken private**.

Typical LBO targets: companies with large **undervalued assets** (sellable to reduce debt) and high, steady **cash flows** (usable to repay debt). The objective is to restructure the company and later **relist** it by issuing new shares in the primary market.

1.6 Investing in foreign (non-domestic) equities

Shares can be issued in markets outside their home country. **Dual-listed** shares are simultaneously issued and traded in two markets. There are two ways in:

- **Direct investing** — buy the foreign share itself on the foreign exchange.
- **Indirect investing** — via **Depository Receipts** and Global Registered Shares.

> **Depository Receipt (DR)** A negotiable instrument issued by a **bank** representing an economic interest in a foreign company, which trades on a domestic exchange just like a common share.

- DRs are **not issued by the company** and raise **no capital** for it — they're issued by financial institutions (banks) to **facilitate trading** of the stock in other countries.
- Transaction costs are **significantly lower** than buying the stock directly on the foreign exchange.
- Known globally as **GDRs** (Global Depository Receipts); in the US as **ADRs** (American Depository Receipts).
- **No voting rights** — those rest with the bank.

> EXAM WATCH Two facts about DRs are favourite questions: they carry **no voting rights**, and they **raise no capital** for the underlying company (the bank issues them, not the company).

1.7 Risk, return and company value

Risk is ever-present in financial markets, and there's a **trade-off between risk and return**. We need to measure both.

Return — the Holding Period Return (HPR)

$$R_t = \frac{(\text{EV} - \text{BV}) + D_t}{\text{BV}}$$

EV = ending value · BV = beginning value · D = dividends received in the period.

Related measures you should know by name: **annualised HPR**, the **arithmetic mean return**, and the **geometric mean return**. Other sources of return beyond price change include **foreign-exchange gains/losses** (for investors holding DRs or foreign shares) and **reinvested dividends**. An asset's **average historical return** can be used as a proxy for its expected future return.

Risk

- The risk of any security is based on the **uncertainty of its future cash flows**.
- The greater that uncertainty, the greater the risk — and the more **variable/volatile** the price.
- Risk is the **uncertainty of expected (future) total return**.
- An asset's average historical return and the **standard deviation** of its returns proxy its expected return and risk.

Company value — market value vs book value

Market value (market capitalisation)

Reflects investors' expectations about the **amount, timing and probability** of the company's future cash flows. Management actions affect it only **indirectly**.

Book value

Based on **historical** values, so it rarely equals market value. It reflects the cumulative effect of management's past actions. **Book value = Total Assets − Total Liabilities.**

Accounting Return on Equity (ROE)

ROE measures whether management is using shareholders' equity **effectively and efficiently** to generate profit. The higher the ROE, the more likely the business can generate cash internally.

$$ROE_t = \frac{NI}{\text{average shareholders' equity}} = \frac{NI}{(SE_t + SE_{t-1}) / 2}$$

WORKED EXAMPLE

FINANCIAL YEAR ENDING	2017	2016
Net income	R10 150 000	R9 995 000
Total shareholders' equity	R49 445 000	R65 500 000

ROE_{2017} = 10 150 000 ÷ [(49 445 000 + 65 500 000) / 2] = **0.1766 = 17.66%**.

ROE can also rise if a company **issues debt to buy back shares**, shrinking the equity base — but this raises **leverage** and makes the equity **riskier**, so a rising ROE is not automatically "good".

Q1. Which statement about preference-share dividends is correct?

A. They are a contractual obligation of the company, like debt interest.
B. They are fixed and generally higher than common dividends, but not a contractual obligation.
C. They are variable and always lower than common dividends.

Q2. A depository receipt gives the holder...

A. voting rights in the foreign company.
B. no voting rights — these rest with the issuing bank.
C. the right to force the foreign company to issue new capital.

Q3. Cumulative voting primarily benefits...

A. majority shareholders.
B. minority shareholders.
C. the company's auditors.

Q4. In a leveraged buyout, the buyers fund the purchase *primarily* with...

A. retained earnings.
B. debt.
C. new common-share issues to the public.

ANSWERS 1-B · 2-B · 3-B · 4-B. Note how each wrong option is either the **opposite** of the truth, a **neighbouring role-player** (auditors), or a plausible-but-wrong funding source — exactly the trap pattern your exam uses.

Chapter 2 — Equity Issuance

Learning goal: understand how shares get to market, who the players are, where they trade, and the rules for listing in South Africa.

2.1 Primary vs secondary markets

Primary market Where **new** shares are issued and the company **raises capital** (e.g. an IPO or a seasoned offering).

Secondary market Where **existing** shares trade between investors. The company raises **no new capital** here — ownership simply changes hands.

LECTURER'S NOTE This is one of the most-tested distinctions in the module. When you buy Naspers shares on the JSE today, Naspers gets nothing — you're in the secondary market. Naspers only received cash at its original primary issuance.

2.2 Market participants & investment positions

The stock market brings together issuers (companies raising capital), investors (individual and institutional), and intermediaries (brokers, dealers, market makers) under the oversight of the exchange and regulators. Investors take one of two basic positions:

- **Long position** — you own the share and profit if the price rises.
- **Short position** — you sell a borrowed share, hoping to buy it back cheaper; you profit if the price falls.

2.3 Listed vs unlisted shares

> **Unlisted equities** Shares **not registered with or traded on a licensed stock exchange**. They trade **over-the-counter (OTC)**, facilitated by market makers or dealers.

Trading unlisted equities is generally **riskier** than listed equities, for several reasons:

RISK	WHY
Lack of regulatory framework	No protection from the Financial Markets Act or a licensed exchange; issuers often can't (or won't) meet stringent listing requirements.
Lack of a formal market	Trades happen directly between buyer and seller — you may struggle to sell because you first have to *find* a buyer.
Pricing & liquidity risk	Often illiquid / "thinly traded", so a robust market price is hard to establish; wide bid-ask spreads; reliance on brokers for prices.
Higher business risk	Many OTC issuers are small/medium companies with limited history, sometimes distressed — investment can lead to total loss.

> **EXAM WATCH** A single market maker often controls an unlisted security's market, so the price can be subject to **manipulation**. Watch for "OTC" being paired wrongly with "fully regulated".

2.4 South African stock exchanges

An exchange is an organised, regulated market that does two core jobs:

- **Match buyers with sellers** — a place where the two sides can quickly and easily find one another.
- **Make sure trades are completed** — once a trade is agreed, the exchange sends the details to a **clearinghouse** to be cleared/finalised.

The JSE is no longer alone. By end-2019 the newer exchanges had more than **R1.9 trillion** listed and saw **47 new listings** over two years vs the JSE's 39. Each has a differentiator:

EXCHANGE	ESTABLISHED / SCALE	DIFFERENTIATOR
JSE	1887; ~354 companies	The primary licensed exchange for IPOs in SA. Settles **T+3**.
ZAR X	Newer	**Same-day settlement** (faster than JSE's T+3); targets unlisted companies.
A2X Markets	Newer	33 **secondary listings** of JSE companies (Naspers, Standard Bank, Sanlam).
4AX	Newer	Targets **unlisted** companies.
EESE	Newer	Strategy: attract **empowerment** companies.

2.5 JSE listing requirements

The JSE is the primary licensed exchange for IPOs, so its rules matter most. It runs **two primary boards**:

- **Main Board** — for established companies.
- **AltX** (Alternative Exchange) — for small and medium-sized companies; eligibility criteria are **significantly less onerous**, so it attracts more development companies.

A company cannot list on the JSE unless the offer complies with the **Companies Act** and the **JSE Listings Requirements** — secondary legislation published by the JSE under the **Financial Markets Act (FMA)**. The guiding principles:

- The JSE must be satisfied the issuer is **suitable** and that listing its shares is appropriate.
- All material activities must be **disclosed promptly** to shareholders and the public.
- Shareholders should receive **full information** and get to **vote** on substantial changes to the business or their rights.
- Those releasing information to the market must observe the **highest standards of care**.
- Holders of the same class of securities must enjoy **fair and equal treatment**.

> FIGURE TO REMEMBER A common AltX-type threshold cited in the material is **subscribed capital of at least R2 000 000** (including reserves but excluding minority interests).

2.6 Methods of listing — IPO, SEO and dual listing

IPO (Initial Public Offering) A company's **first** sale of shares to the public in the primary market. It includes a **price-setting** process.

SEO (Seasoned Equity Offering) A company that is **already listed** issues more shares. It follows many of the same steps as an IPO, but the key difference is that the **price-setting process is not necessary** (a market price already exists).

Dual listing

Dual listing is listing a security on **two or more exchanges**. A security can list on any exchange whose requirements it meets and whose fees it pays. Benefits: **more liquidity** (more venues, participants, sometimes more hours), **more capital** and **exposure** (wider investor base). A popular form is through **depository receipts**.

SA examples: BHP Billiton, AB InBev, Richemont, Naspers, Anglo American.

2.7 Share prices, indices & common stock

How share prices are set

By **supply and demand**, reflected in the **bid-ask spread**. More buyers than sellers → price rises; lots of sellers and no buyers → price falls quickly.

Common stock share

A claim to a share of a corporation's **assets** and its **net income** (income after taxes, expenses and debt). So the return depends on the **economic performance** of the issuer.

All Share Index — a benchmark tracking the broad market; widely reported, which is why even non-investors are aware of "the market".

Q1. When an investor buys existing shares on the JSE, the issuing company receives...

A. the full purchase price as new capital.
B. nothing — it is a secondary-market trade.
C. a listing fee rebate.

Q2. The main difference between an SEO and an IPO is that an SEO...

A. does not require a price-setting process.
B. can only be done on the AltX.
C. raises no capital for the company.

Q3. Which SA exchange is known for same-day settlement of share trades?

A. A2X Markets.
B. ZAR X.
C. EESE.

Q4. Unlisted equities are riskier mainly because... *(choose the best answer)*

A. they are over-regulated by the FMA.
B. they trade OTC with no formal market and poor liquidity.
C. they always pay no dividends.

ANSWERS 1-B · 2-A · 3-B · 4-B. Q3's distractors are **adjacent role-players** (other real exchanges); Q4-A is the **opposite** of the truth (OTC is *under*-regulated).

Chapter 3 — Equity Valuation: Concepts & Basic Tools

Learning goal: understand why we value equity, the difference between intrinsic value and market price, and the families of models used to estimate value.

3.1 The starting assumption — going concern vs liquidation

Before you value anything you must decide which assumption applies:

Going concern

- The firm **continues** its business activities.
- Continues to **sell its goods and services**.
- Uses its assets for **value maximisation**.
- Accesses its **optimal sources of financing**.

Liquidation

- The firm will be **dissolved**.
- The firm's assets will be **sold separately**.

3.2 Intrinsic value vs market price

Intrinsic value The underlying, essential, inherent value of an asset given a **complete understanding** of its investment characteristics — based on **quantitative** factors (capital, earnings, revenue) and **qualitative** factors (management quality, intellectual capital, track record). Because the factors weighed differ from investor to investor, intrinsic value estimates differ too.

IV > MV	Asset is undervalued → a buy
IV < MV	Asset is overvalued → avoid / sell

EXAM WATCH The direction trips people up. Higher intrinsic value than market price means the market is **underpricing** it → **undervalued**. Don't reverse it.

3.3 Why we value equity — the uses of valuation

USE	THE QUESTION IT ANSWERS
Security selection	Is a share undervalued or overvalued?
Inferring market expectations	What does the market price imply about investors' expectations of the firm?
Evaluating corporate events	What's the effect on firm value of an **acquisition** (a merger where one firm is the acquirer), a **divestiture** (selling a major business component), a **spin-off** (separating a component and transferring ownership to shareholders), or a **leveraged buyout**?
Fairness	Does the firm value / share price represent fair value?
Evaluating business strategies	What's the value effect of a new strategy?
Communicating	With analysts and shareholders — how is value affected?
Appraising private businesses	What is the value of the private firm?
Compensation	What is the value of equity compensation?

3.4 The valuation process

Top-down approach

Start broad and narrow down: **economy** → **industry/sector** → **company**.

Bottom-up approach

Start with the **company's own fundamentals** and build up.

The process runs: **understand the business** → forecast company performance → **select a valuation model** → convert forecasts into a valuation → apply the conclusions. The first step — *understand the business* — underpins everything: you cannot value what you do not understand.

3.5 Selecting a valuation model

1 · Absolute valuation (Discounted Cash Flow) models

Value the asset on its **own** expected cash flows, discounted to present value:

- **Dividend Discount Model (DDM)** — PV of future dividends.
- **Free Cash Flow to Equity (FCFE)** — PV of future cash flows available to shareholders *after* debt is paid.
- **Free Cash Flow to the Firm (FCFF)** — PV of future cash flows available to the firm.
- **Residual Income** — PV of future net income in excess of the required return on equity.

2 · Relative valuation

- **Price ratios** — **P/E**, **P/B**, **P/S**.
- **Enterprise value multiples** — e.g. **EV/EBITDA**.

3 · Asset-based

NAV / FMV = total assets − total liabilities. Best for firms with many tangible assets and **few intangibles**.

Choosing the right model

CONSIDERATION	GUIDANCE
Company characteristics	Absolute valuation requires forecastable **future cash flows**; asset-based valuation requires significant **tangible assets**.
Data availability & quality	**DDM** needs a company that **pays dividends** (stable, predictable, mature); a **relative P/E** needs **positive earnings**; **FCFF/FCFE** suit firms with no/irregular dividends but positive, predictable free cash flows.
Purpose of the valuation	**FCF** approaches suit **controlling** ownership; **DCF** can be useful for a **minority** share ownership perspective.

LECTURER'S NOTE The model must fit the company. Using a DDM on a company that pays no dividend, or a P/E on a loss-making company, is the classic beginner error. Match the tool to the data you actually have.

CHAPTER 3 · SELF-TEST

Q1. If a share's intrinsic value is greater than its market price, the share is...

A. overvalued.
B. undervalued.
C. fairly valued by definition.

Q2. Which model best suits a mature company with a stable, predictable dividend history?

A. Asset-based / NAV model.
B. Dividend Discount Model (DDM).
C. EV/EBITDA multiple.

Q3. A firm separates one of its components and transfers ownership to its shareholders. This is a...

A. divestiture.
B. spin-off.
C. acquisition.

ANSWERS 1-B · 2-B · 3-B. A **spin-off** transfers ownership to *shareholders*, whereas a divestiture *sells* a component to an outside buyer.

Must-know definitions

- Equity capital = share capital (ownership).
- Capital structure = the debt/equity mix.
- Residual claimant = common shareholders rank last.
- DR = bank-issued claim on a foreign share; no votes, no capital raised.
- Primary = new capital; Secondary = existing shares.
- IV > MV = undervalued.

Must-know formulas

HPR: $R = [(EV - BV) + D] / BV$

ROE: $NI \div [(SE_t + SE_{t-1})/2]$

Book value: Total Assets − Total Liabilities

NAV: Total Assets − Total Liabilities (asset-based value)

Must-know distinctions

- **Common vs preferred:** votes & residual claim vs fixed (non-obligatory) dividend & priority.
- **Statutory vs cumulative voting:** per-director vs pooled (protects minorities).
- **Cumulative vs participating preferred:** missed dividends accrue vs share in extra profit/assets.
- **IPO vs SEO:** first issue with price-setting vs follow-on with no price-setting.
- **Going concern vs liquidation:** keeps trading vs dissolved & assets sold separately.
- **Absolute vs relative vs asset-based:** own cash flows vs peer multiples vs net assets.